

2D Videogame called True Name

Filip Dravecký

ČVUT-FIT

dravefil@fit.cvut.cz

January 4, 2026

1 Introduction

I have created a simple 2D videogame using the Python programming language and the Pygames module. It is, utilizing the pretrained Pytorch model for letter recognition. My goal was creating a seamless game loop, where player is rewarded for his effort.

2 User input

User is navigating by clicking onto buttons. Gameplay is forcing player to readable, space efficient writing and pattern matching.

3 Methods

I have created a short and clean main file which only handles the game loop. All other parts of the code are handled separately. I have used object oriented programming so most of the code is in separate classes. It is easy to read and expand furthermore. Lastly for letter matching I have pretrained small Pytorch model on small scale training dataset.

4 Results

I have accomplished that player is able to store and reset his progress as he pleases.



Figure 1: *Saved data*

Next thing, that I have accomplished is seamless gameloop. Where player is rewarded for his effort.



Figure 2: *Seamless gameloop*

Last but not least, my image to letter evaluation system is quite accurate.



Figure 3: *Evaluating system*

5 Conclusion

This project taught me lots of new things, helped me on getting to know python better, gave me more insight on game development. Also I improved my understanding of user experience.

References

- [1] Universität Leipzig. Downloads for english – wortschatz leipzig. online, 2025. [cit. 2025–10–12] <https://wortschatz.uni-leipzig.de/en/download/eng>.
- [2] Sujay Mann. Handwritten english characters and digits. online, 2025. [cit. 2025–14–12] <https://www.kaggle.com/datasets/sujaymann/handwritten-english-characters-and-digits>.
- [3] Timotej Molnár. Wizzard asset. online, 2025. [cit. 2025–10–12] https://www.instagram.com/uncle_wizzard/.
- [4] Rob Mulla. Build your first pytorch model in minutes! [tutorial + code]. online, 2024.

[cit. 2025-14-12] <https://www.youtube.com/watch?v=tHL5STNJKag>.

[5] osadam. Gothic pixel art font. online, 2025. [cit. 2025-10-12] <https://osadam.itch.io/gothic-pixelart-font>.

[6] Bratislava: Jazykovedný ústav Ľ. Štúra SAV. Slovenský národný korpus – prim-8.0-public-sane. online, 2018. [cit. 2025-10-12] <https://korpus.juls.savba.sk>.